In silico project(ver3.5.1)
User Manual
for Window

<Module>

NGS dataset downloader(NDD)

In silico sequence mining(ISSM)

https://github.com/khk1329/NDD-and-ISSM.git

If you encounter any issues or have suggestions for improvement,

please contact us at the email address below.

We appreciate your feedback.

kminchan1010@naver.com

ghkrk007512@naver.com

Introduction

- The GUI-based in silico project provides two selectable modules.

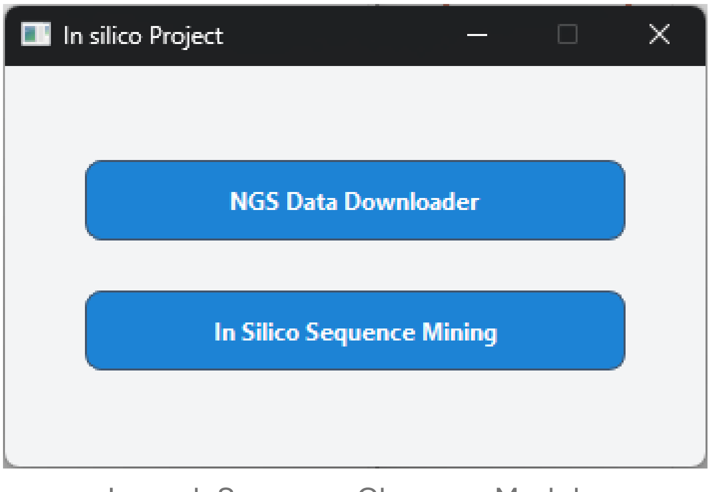
1) NDD (NGS Data Downloader) enables users to search and easily download datasets from the most authoritative repositories, NCBI and ENA.
2) ISSM (In Silico Sequence Mining) analyzes sequence-based datasets(FASTA, FASTQ) to detect sequences that match a specified target probe. In addition to reporting match presence or absence, ISSM also extracts and presents the matched sequences for verification.

Downloading and Launching

- After downloading the .exe (executable) file, you can launch the program directly by double-clicking it. No additional installation or configuration is required.

How to use the tool

- When you run the .exe file, a graphical user interface (GUI) will open. From this interface, you can choose between the two available modules: NDD and ISSM



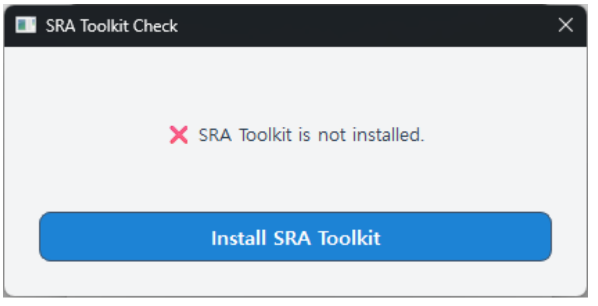
Launch Screen — Choose a Module

How to use the tool - NDD

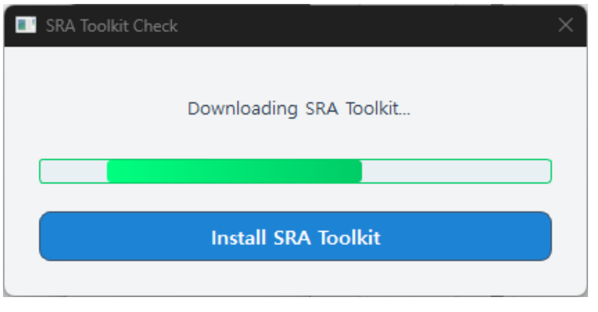
- **Installation & Check of SRA Toolkit**

When the program is first launched, it automatically checks whether the SRA Toolkit is installed on your computer.If the toolkit is not found, the program will guide you through an automatic installation and configure the environment variables.

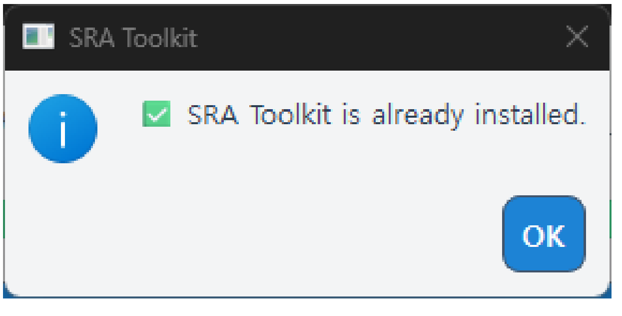
Once the process is complete, the toolkit is ready for immediate use.



If not detected, click Install SRA Toolkit to install.



Downloading and Setting SRA Toolkit



Completed message

How to use the tool - NDD

- **Main window of NDD**

You can select the database from the top-left of the window.

Each GUI differs slightly in layout, but they operate in the same overall workflow. Depending on the selected database, you can narrow down the search results by using keywords, just as you would on the official website.

SRA (NCBI):

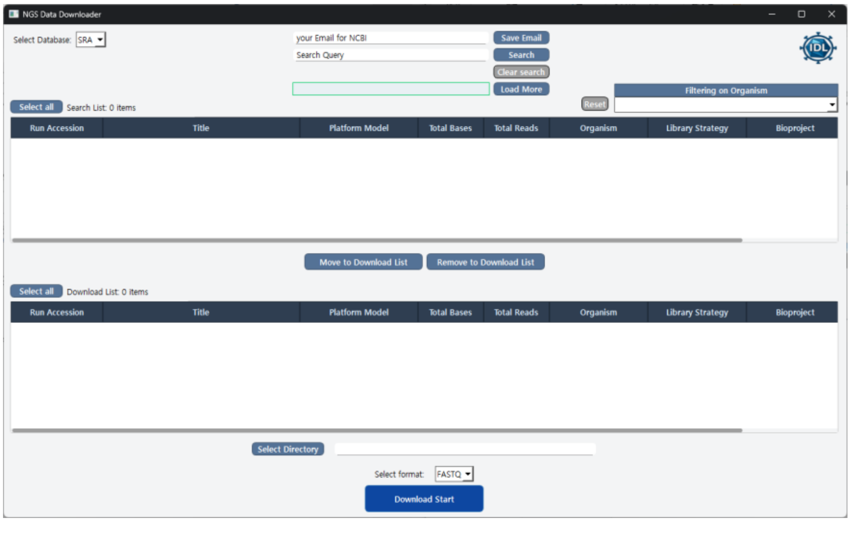
Due to NCBI limits, a single query can return up to 9,999 results at a time. To retrieve additional results for the same query, use Load More.

Query ex) ACE2[Gene] AND "Eptesicus serotinus"[Organism] AND "RNA-Seq"[Strategy]

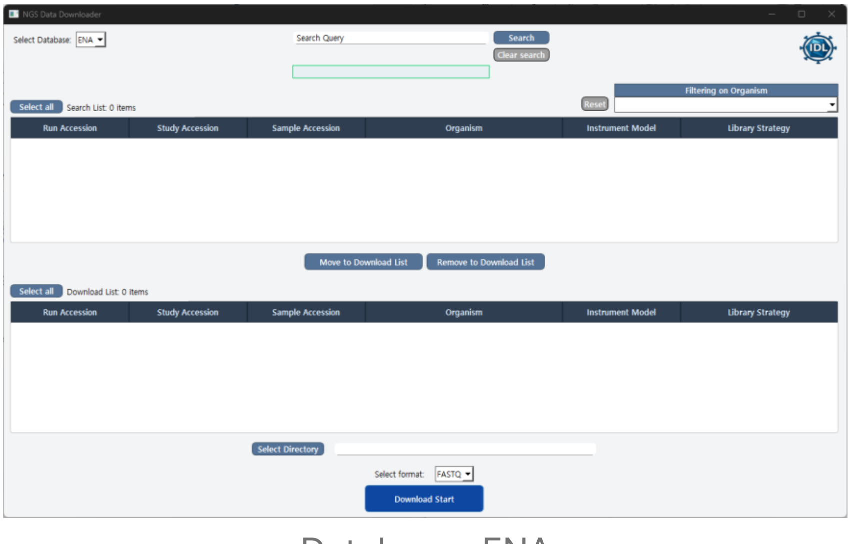
ENA:

All matching results can be fetched in one request, so a Load More button is not required

Query ex) scientific_name="Eptesicus serotinus" AND (description="ACE2" OR study_title="ACE2")



Database : SRA

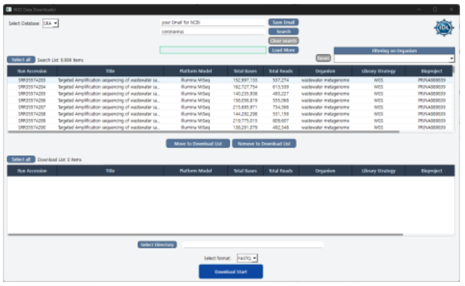


Database : ENA

How to use the tool - NDD

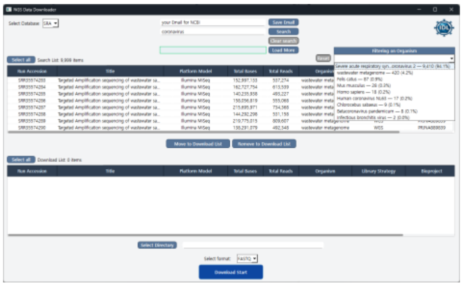
- **Process of SRA download**

1. **Enter Email & Search**
   For stable access to NCBI, enter your email address, then type your query and click Search.

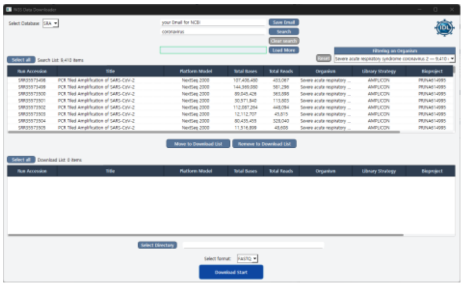


After searching — Results displayed in the GUI

2. **Filter by Organism**
   Review the organism distribution in Filtering on Organism (right panel). Click an organism to view the filtered results.



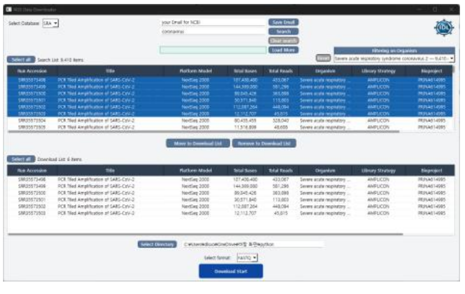
Organism distribution
— Summary of organisms in the results.



Filtered on organism
— Results after selecting an organism.

3. **Select Items & Build the Download List**
   From the search results, select the runs you want and click Move to Download List to add them to the list below.

4. **Set Output Folder**
   Specify the output folder where downloaded files will be saved.



Moved to download list — Selected runs added to the download list.

How to use the tool - NDD

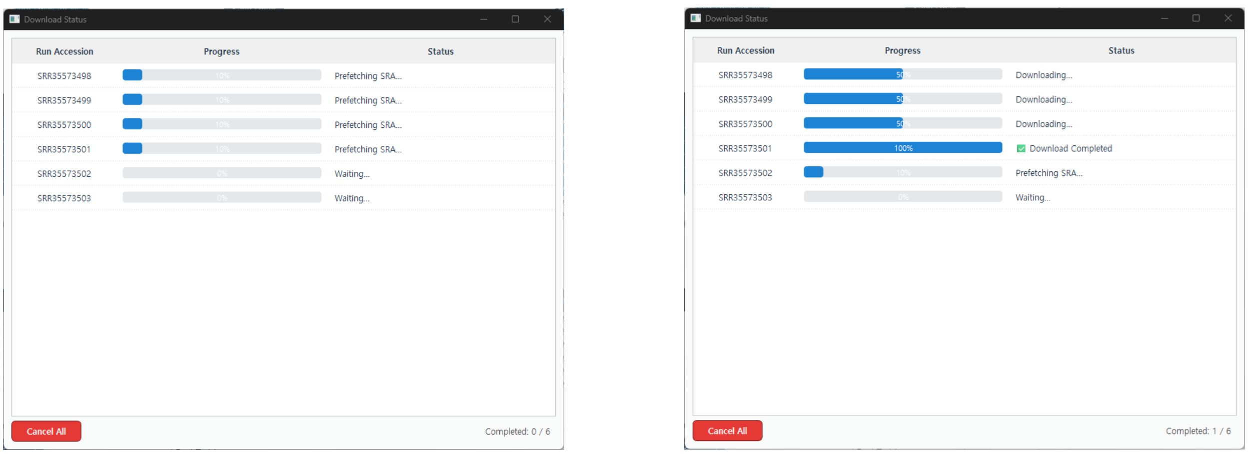
- **Process of SRA download**

5. Start Download

Click Download Start to download all items in the download list.

6. Monitor Downloads

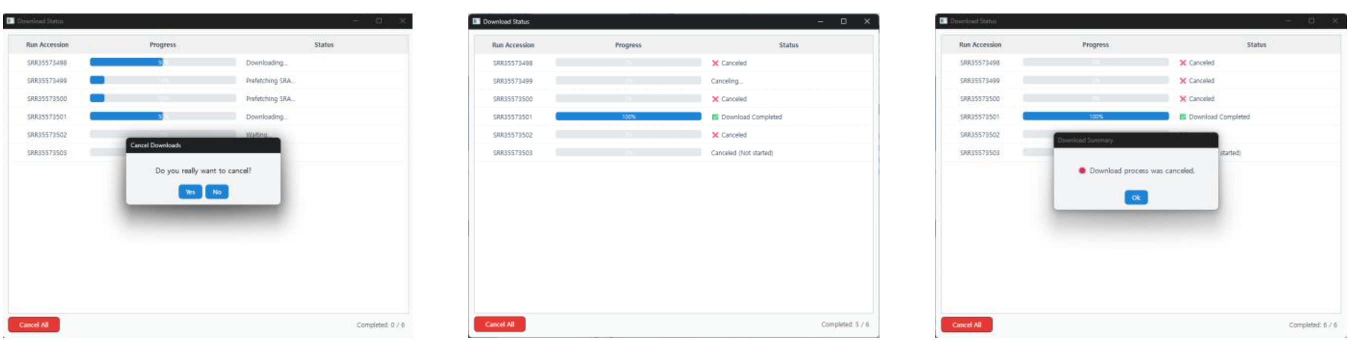
You can track the progress in real time through the Download Status window.



Download status — View the progress of each run during download.

*If you want to cancel during a download, click Cancel All (bottom-left) to stop ongoing downloads, cancel queued items, and clean up temporary files.

(Completed files remain in the output folder.)



Canceling process — Stops all pending downloads and removes temporary files.

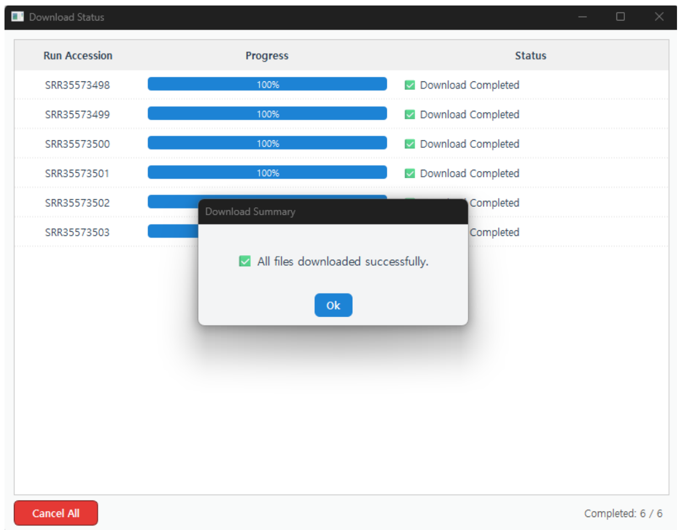
How to use the tool - NDD

- **Process of SRA download**

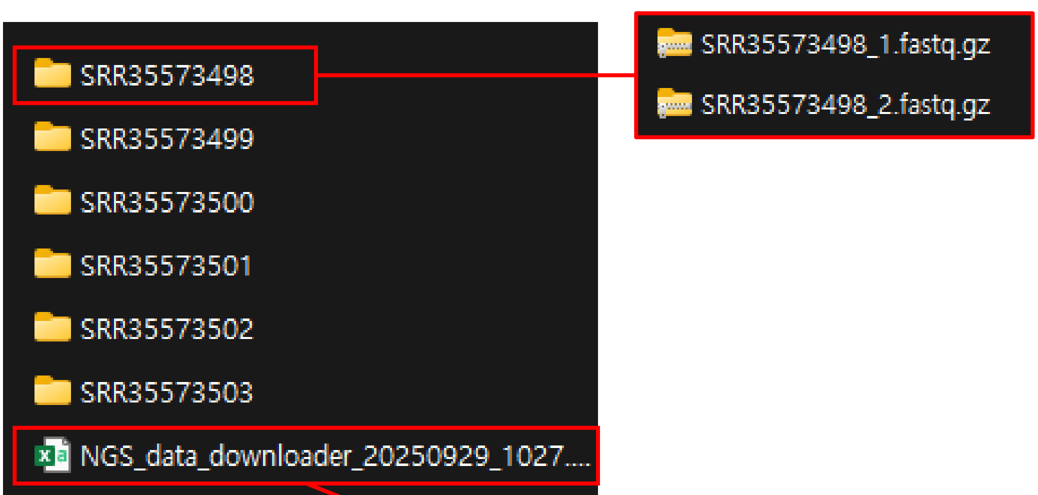
7. Download completed

When all files are downloaded, a completion message is displayed.

The chosen output folder contains a subdirectory for each run with the corresponding FASTQ/FASTA files, along with a CSV log file that records file details and download time.



Download completed message



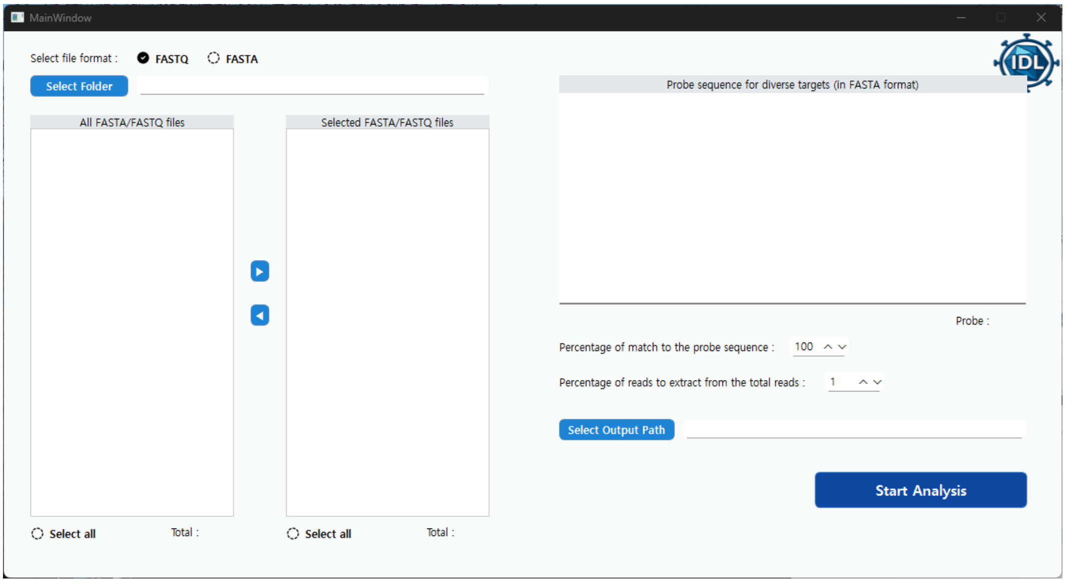
Run Accession	Title	Platform	Bases	Reads	Organism	Strategy	Bioproject	Biosample	Start Time	End Time	Elapsed Time	Size (MB)
SRR35573501	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	30,571,840	113,803	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796698	2025-09-29 10:26	2025-09-29 10:27	0:00:26	14.24 MB
SRR35573500	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	89,045,426	363,898	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796699	2025-09-29 10:26	2025-09-29 10:27	0:00:50	45.78 MB
SRR35573499	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	144,369,080	581,296	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796619	2025-09-29 10:26	2025-09-29 10:27	0:00:59	70.93 MB
SRR35573498	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	107,408,480	433,067	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796700	2025-09-29 10:26	2025-09-29 10:27	0:00:59	54.71 MB
SRR35573503	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	12,112,707	45,815	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796696	2025-09-29 10:27	2025-09-29 10:27	0:00:15	3.80 MB
SRR35573502	PCR Tiled Amplification of SARS-CoV-2	NextSeq 2000	112,087,264	448,094	Severe acute respiratory syndrome coronavirus 2	AMPLICON	PRJNA614995	SAMN51796697	2025-09-29 10:27	2025-09-29 10:27	0:00:44	56.47 MB

Output folder structure

How to use the tool - ISSM

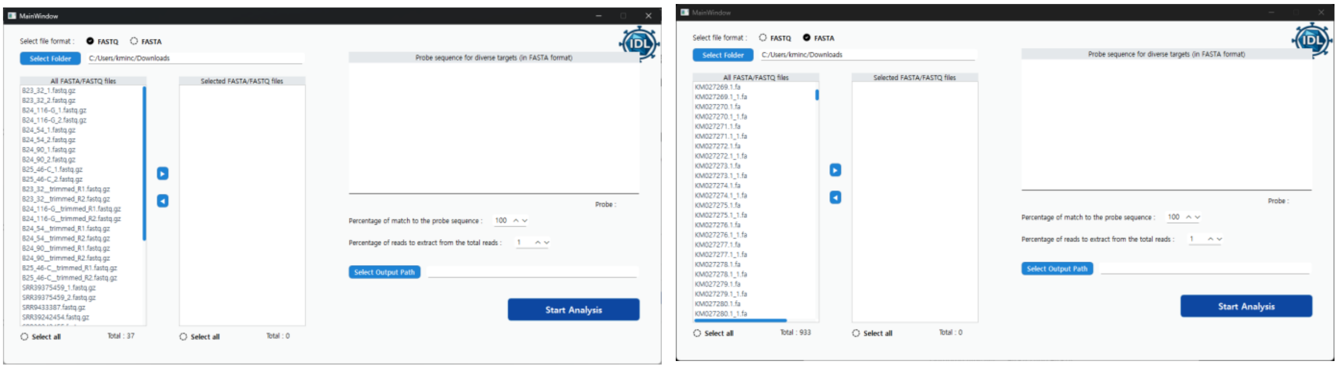
- **Process of ISSM**

1. **Load Main Window**



Main widnow of ISSM

2. **Select File Format and Folder**
   Select the file format (FASTA or FASTQ) and specify the folder where the files are located. A list of files in the selected format found in the chosen folder and all of its subfolders will be displayed.



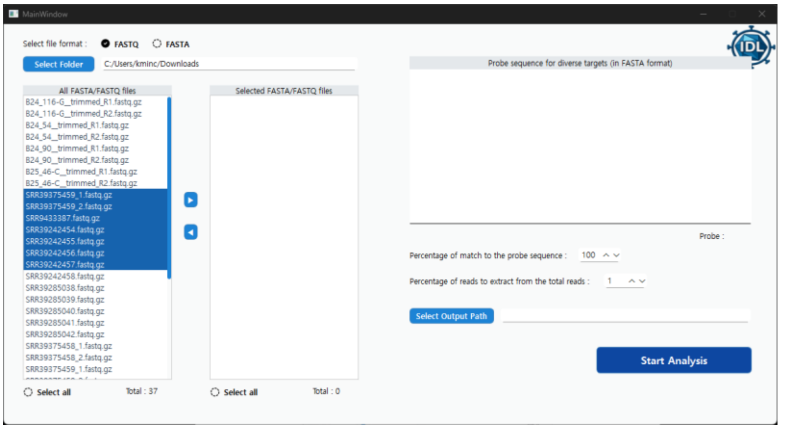
Detected files list — FASTQ and FASTA

How to use the tool - ISSM

- **Process of ISSM**

3. Move to Selected Files List

Select the files to analyze, then use the arrow button to move them to the Selected Files list on the right. (You can multi-select with Click)

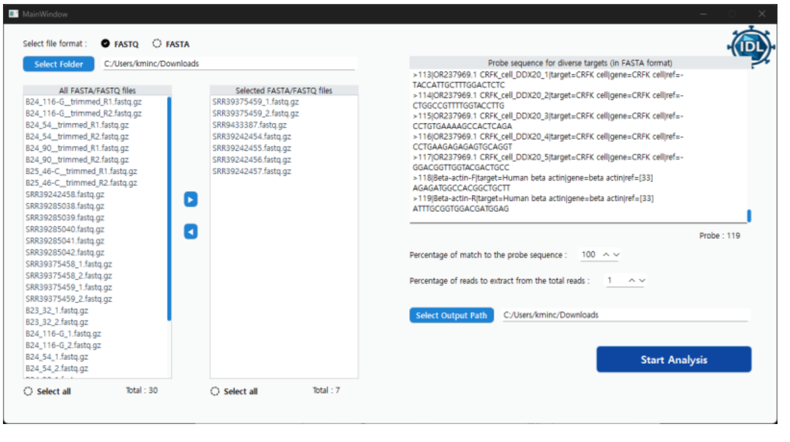


Selected items from All Files list

4. Set Analysis Parameters

Enter the target probe sequence(s) in FASTA format.

Set the match threshold (percent identity to the probe), choose the sampling rate (percent of total reads to analyze), and specify the output path for saving results.

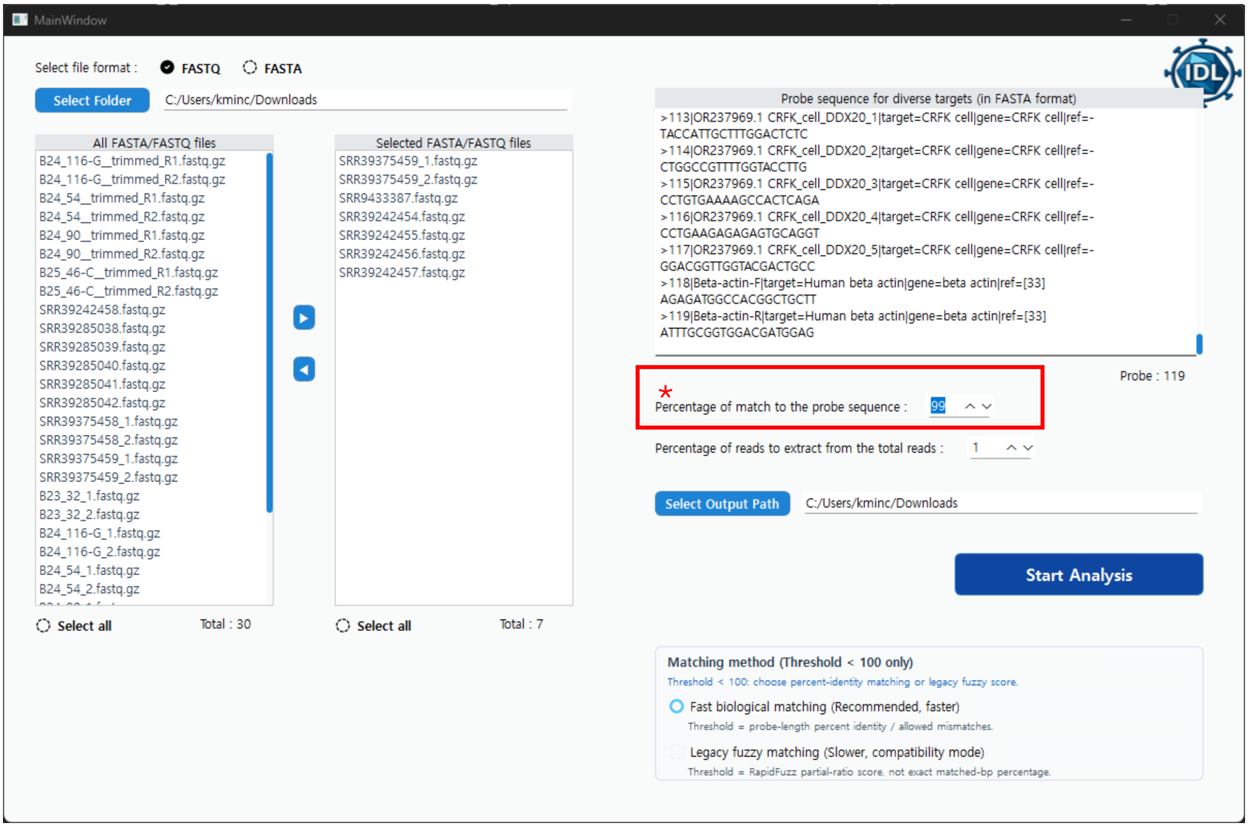


Set All Parameters

How to use the tool - ISSM

- **Process of ISSM**

4. Set Analysis Parameters



When Percentage of match to the probe sequence is set below 100, the Matching method panel is displayed. When the value is set to 100, exact matching is applied automatically.

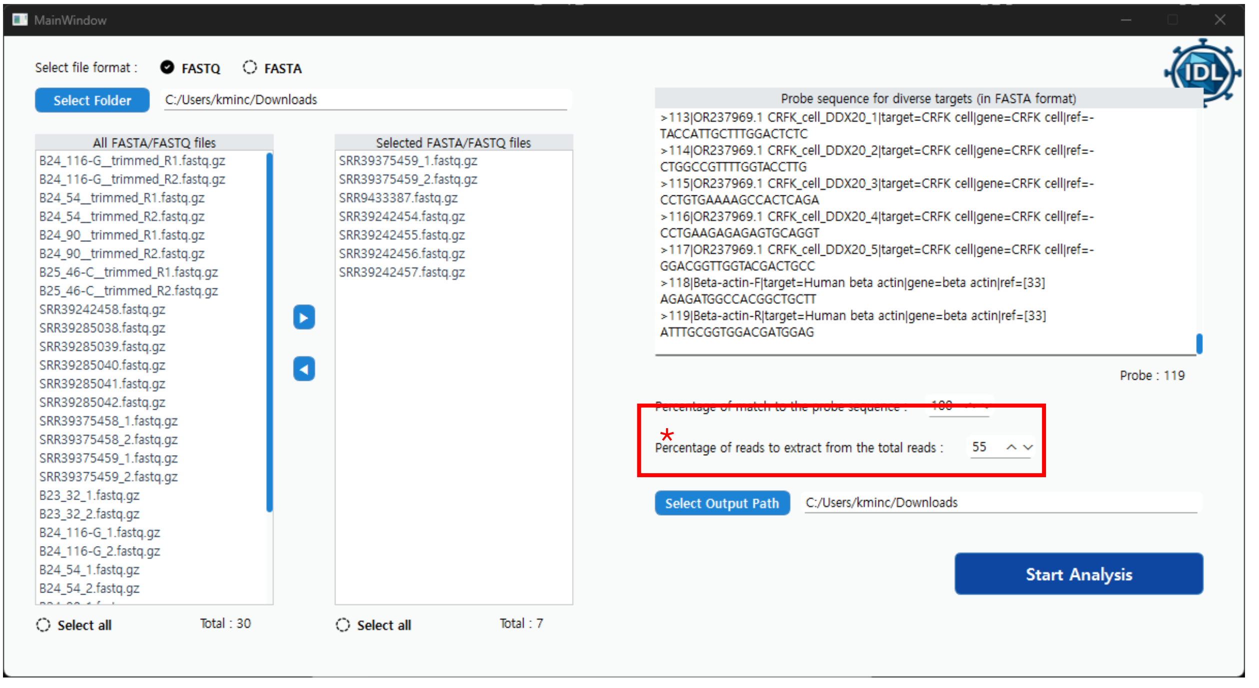
Fast biological matching is the recommended method. It compares input sequence windows of the same length as the probe and detects matches according to the allowed number of mismatches. For example, for a 20 bp probe with Percentage of match to the probe sequence set to 95%, up to one mismatch is allowed, and the sequence is detected when at least 19 bp match. This method uses a Rust-based matcher and is therefore fast, but it does not allow insertions or deletions.

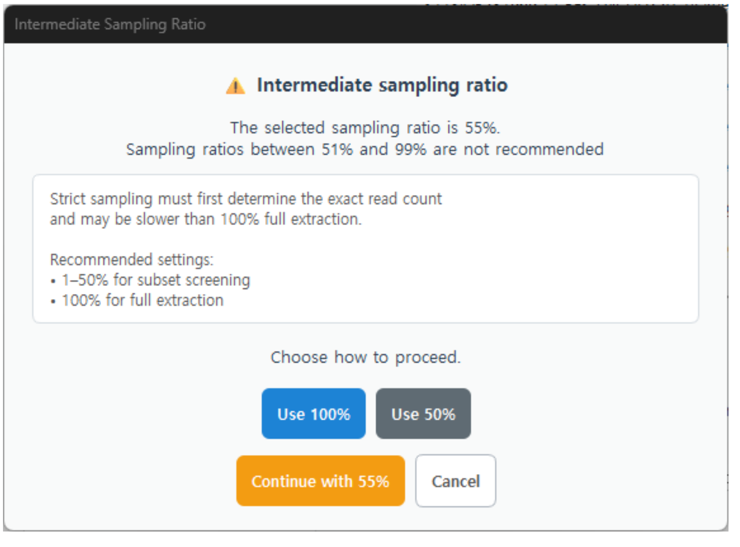
Legacy fuzzy matching uses the original ISSM matching method and determines matches based on the RapidFuzz similarity score. A sequence is detected when the calculated score is equal to or higher than the user-defined Percentage of match to the probe sequence value. This score is not identical to the actual percentage of matching base pairs. This method is intended for compatibility with previous ISSM results and may be slower than Fast biological matching.

How to use the tool - ISSM

- **Process of ISSM**

4. Set Analysis Parameters





When Sampling percentage is set between 51% and 99%, an Intermediate Sampling Ratio warning message is displayed. Strict sampling requires the program to count all input records before generating an exact-sized random subset, so analysis at these ratios may take longer than analyzing 100% of the input data.

Users can change the sampling percentage to 50% or 100%, continue with the current value using the advanced option, or cancel the analysis. For efficient screening, 1–50% is recommended for subset analysis, while 100% is recommended for full analysis.

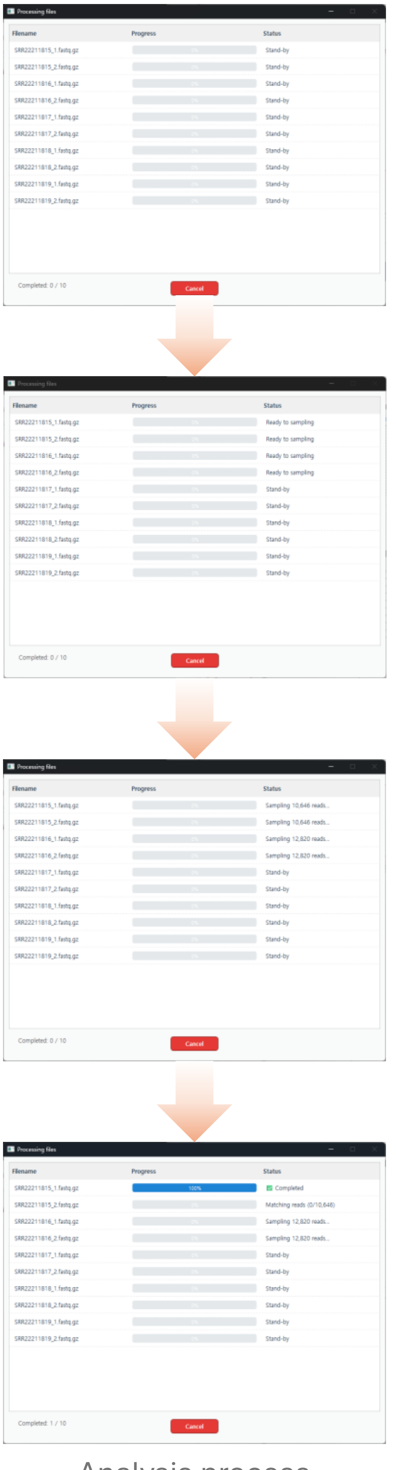
How to use the tool - ISSM

- **Process of ISSM**

5. Run Analysis and Monitor Analysis

The process window displays real-time sampling and analysis status for each file.

When the extraction percentage is below 100, sampling is performed; for FASTQ, the sampled reads are saved as a temporary .fastq.gz and then analyzed.



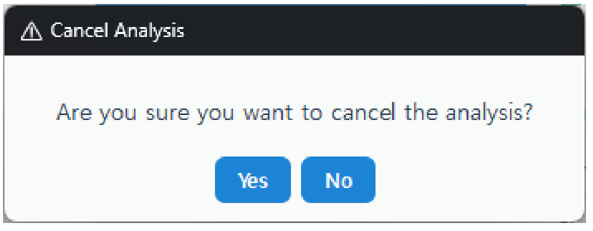
Analysis process

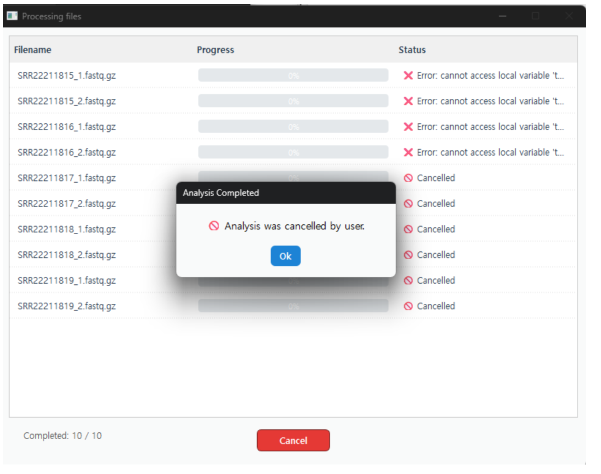
How to use the tool - ISSM

- **Process of ISSM**

5. Run Analysis and Monitor Analysis

*If you want to cancel during analysis, click Cancel All (bottom-middle) to stop all queued and in-progress processes and confirm cancellation for all files.

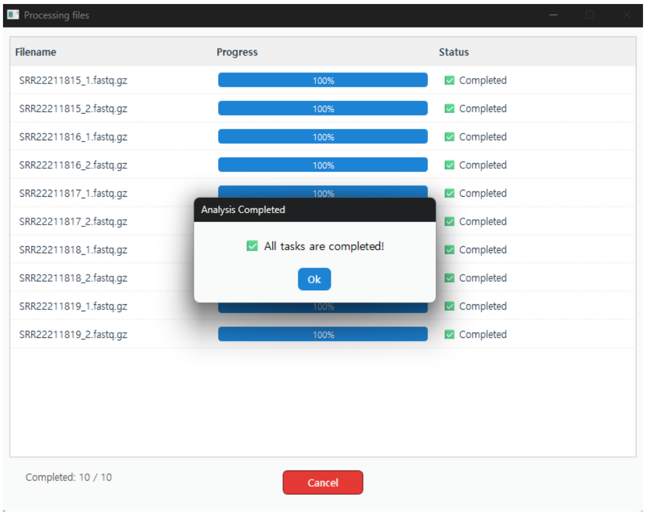




Canceling process — Stops all pending analysis

6. Analysis Completed

When analysis of all files is complete, a completion message is displayed.



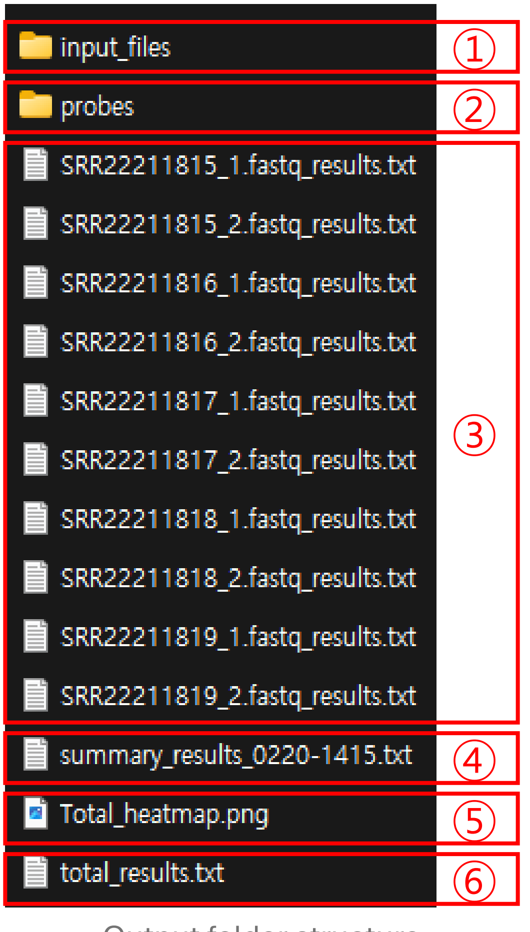
Download completed message

How to use the tool - ISSM

- **Process of ISSM**

6. **Analysis Completed**
   In the chosen output folder, you will find the data for each file along with a combined summary file for all analyses.

   ① Input_files folder :
   Per-file plots showing how multiple probes matched (x-axis = probes)
   ② Probes folder :
   Per-probe plots showing how multiple input files matched (x-axis = input files)
   ③ input file name_results.txt:
   Results for each individual file
   ④ summary_results_start-time.txt:
   Number of reads(seqeuences) analyzed and processing time
   ⑤ Total_heatmap.png :
   Heatmap of matched results across all files
   ⑥ total_results.txt:
   Matched read(sequences) counts and percentages for each file



Output folder structure

NDD & ISSM Troubleshooting

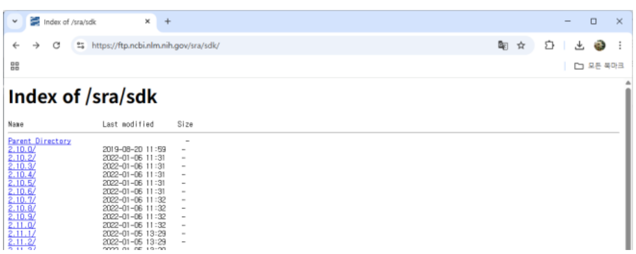
1. If the SRA toolkit is not detected

- **Cause**

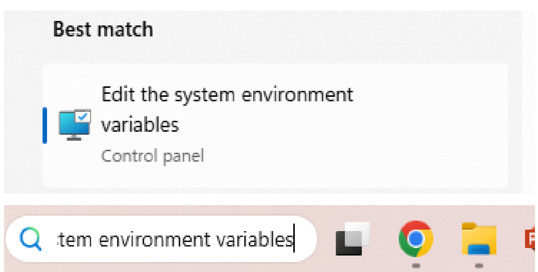
- If the issue is not related to internet connection or insufficient disk space, it is highly likely caused by the SRA toolkit folder not being included in the system environment variables.
- Even if the folder is included in the system environment variables, the problem may occur if Windows does not refresh automatically.
- The issue may also occur if the SRA toolkit was not properly downloaded for other reasons.

- **Solution**

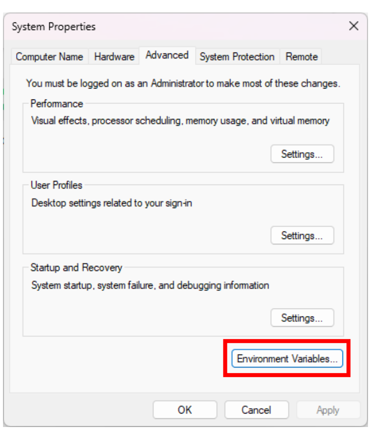
- Verify that the internet connection is stable and there is sufficient disk space.
- Check whether the SRA toolkit folder has been downloaded and confirm that the folder is included in the system environment variables.
- If the path exists in the system environment variables, restart the computer and try again.
- If the toolkit is still not recognized after following the above steps, proceed with the instructions below.

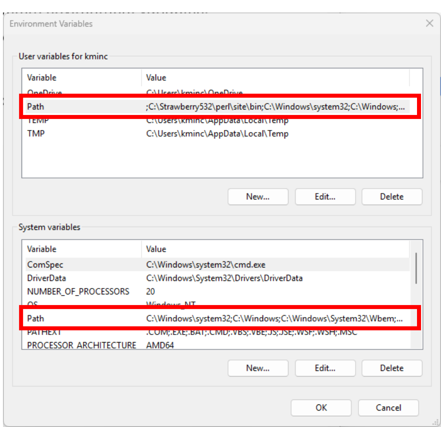


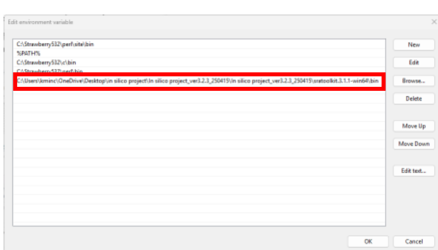
Access the site https://ftp.ncbi.nlm.nih.gov/sra/sdk/ and download the desired version of the SRA toolkit.



Search for and run Edit the system environment variables.







Add the folder path of the SRA toolkit following the order shown in the picture.

NDD & ISSM Troubleshooting

2. Unresponsive behavior during search and filtering in NDD

- **Cause**

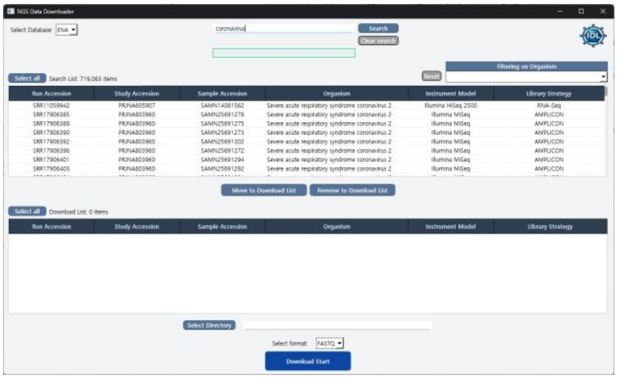
- The issue occurs due to the processing of a large amount of data within the program caused by excessive search results.
- In SRA, the search result is limited to 9,999 entries at a time, so this issue is less noticeable. However, in the case of ENA, all results are returned due to its search logic, making this issue more likely to occur.

- **Solution**

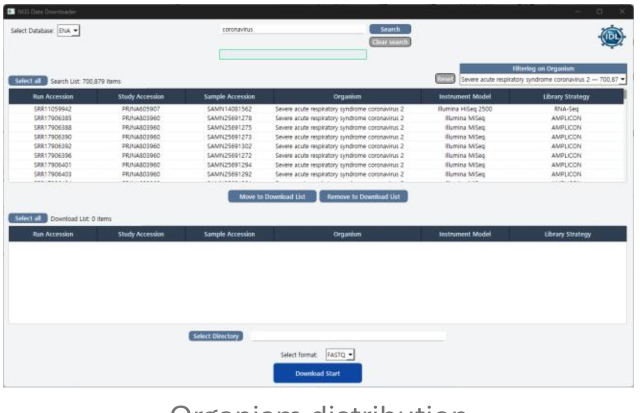
- Although the system may appear unresponsive, the search and filtering processes are still running in the background, and the operation will complete once processing is finished.
- Since the problem arises from an excessive number of search results, please narrow down the search by following the methods suggested in the guide to reduce the size of the result list.

SRA Query ex) ACE2[Gene] AND "Eptesicus serotinus"[Organism] AND "RNA-Seq"[Strategy]

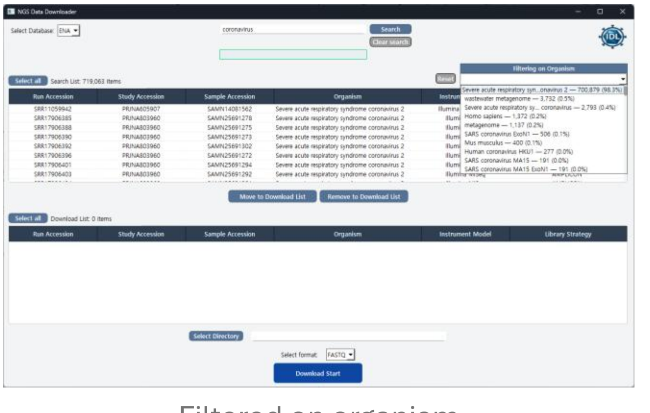
ENA Query ex) scientific_name="Eptesicus serotinus" AND (description="ACE2" OR study_title="ACE2")



After searching — Results displayed in the GUI



Organism distribution
— Summary of organisms in the results.



Filtered on organism
— Results after selecting an organism.

NDD & ISSM Troubleshooting

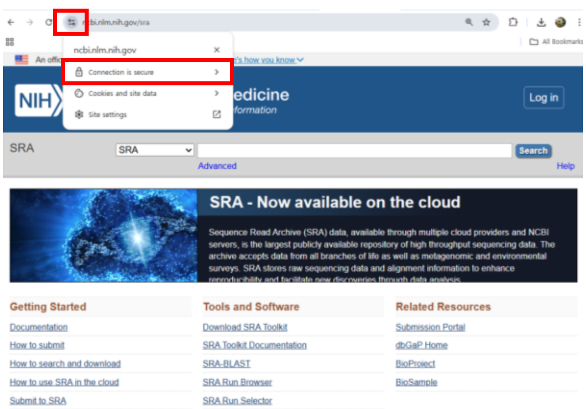
3. In case of a certificate error (Search Failed: FAILED] certificate verify failed: self-signed)

- **Cause**

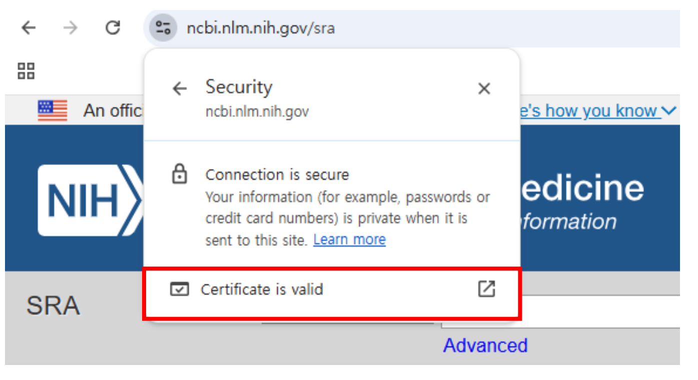
- This issue occurs because the certificate is not automatically retrieved from SRA or ENA.

- **Solution**

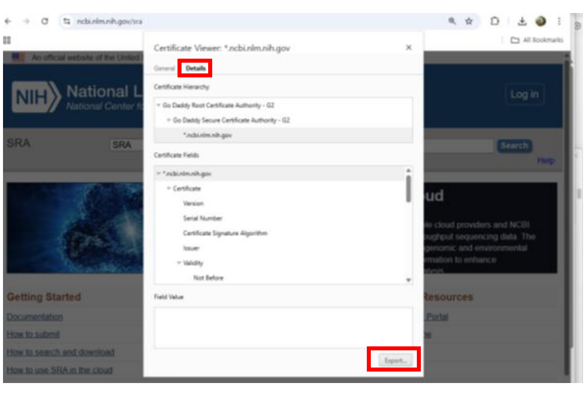
- Manually retrieve and register the certificate by accessing the site directly.
- Detailed instructions are provided below.



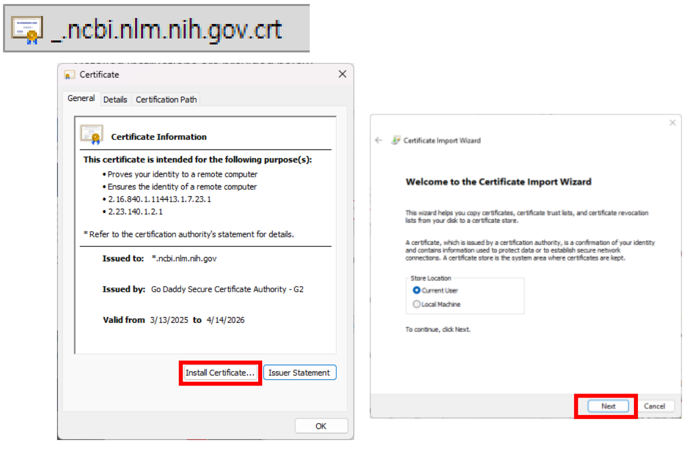
Go to the SRA & ENA website and click on Connection is secure (in Chrome).



Click on Certificate is valid (in Chrome).



Click on Export (in Chrome)



Run the downloaded certificate file and proceed with the certificate installation.